



Zachary Treisman

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Academic Positions

- **Faculty & Honors Program Director, Western Colorado University.**
 - Primary academic statistician on campus; teach 8–15 credits per semester across the mathematics curriculum from Quantitative Literacy through graduate-level coursework.
 - Direct an interdisciplinary honors program of ~150 students, managing a cross-disciplinary faculty team; led redesign of Western's General Education program.
- **Postdoctoral Fellow, University of British Columbia.**
- **Visiting Lecturer, University of Washington.**
- **Visiting Fellow, Tata Institute of Fundamental Research.**

Teaching

- **Western Colorado University:** 8–15 credits per semester. Classes include:
 - Abstract Algebra
 - Advanced Ecological Data Analysis
 - Analysis
 - Calculus I
 - Calculus I (Honors)
 - Calculus III
 - College Algebra
 - History of Mathematics
 - Honors: The Shape of Space
 - Intro. to Advanced Mathematics
 - Intro. to Complex Analysis
 - Precalculus
 - Probability and Statistics
 - Quantitative Literacy
 - Mathematics Senior Seminar
 - Special Topics: Dimension
 - Statistical Modeling and Simulation
 - Topics in Geometry
- **SIMUW** (Summer Institute for Mathematics, University of Washington): Mini-courses on **The Shape of Space**, **The Hopf Fibration**, and **Non-Euclidean Geometry**.
- **UBC:** Calculus III; Linear Programming.
- **University of Washington:** Advanced Multivariable Calculus; Business Algebra; Business Calculus; Calculus I; Geometry for High School Teachers; Introduction to Differential Equations; Linear Algebra; Linear Analysis.

Grants

- **NRT-HDR: Convergent Research in Integrative and Public Land Management.** *NSF 21-536 National Science Foundation Research Traineeship (NRT) Program.* Co-PI.

University Service

- General Education Committee
- Headwaters/Delta Task Force
- Core Liberal Arts & Sciences Task Force
- Curriculum Committee
- Honors Council
- Mathematics work-study coordinator
- Senior seminar mentoring
- IT Committee
- Assessment Committee

Invited Talks (Selected)

- **The vague boundary between geometric models and mathematical art.** Joint Mathematics Meetings, Denver.
- **To the World's End/ A Circle Bundle Over a Circle.** Bridges Stockholm 2018.
- **Mapmaking and vice-versa.** Adams State University; Reed College; Shift Studio Gallery *Dialogues with Artists and Mathematicians*, Seattle, WA.
- **Image and pattern in hyperbolic geometry.** Plenary lecture, Mathday, University of Washington; Adams State College; Colorado College; Western State College.

- **Rationally generated arcs.** University of British Columbia; Institute of Mathematical Sciences, Chennai, India.
- **Arc spaces and rational curves.** Freie Universität, Berlin, Germany.

Art and Mathematics

- *Rule of Law.* A section of a Hopf fibration. With London Tsai.
<http://www.londontsai.com>
- *To the World's End.* A circle bundle over a circle. With London Tsai.
Frontispiece, Loring Tu, *Differential Geometry*. <http://www.londontsai.com/qr/worldsend.htm>
- *Hyperbolic Immersion.* A physical model of non-Euclidean geometry. Slated for installation near Walsenburg, CO.
- *The Hopf map, a flipbook.* An animation of a Hopf fibration. With Matt Kahle.
- *Digital Sundial.* With Jordan Schwartz.

Selected Publications

- **The Boundary of the Visible.** *TSAIBERNETICS: transgenerational cybernetic art* — Wen-Ying Tsai, London Tsai, Ming Tsai. Laznia Centre for Contemporary Art.
- **To the World's End/ A Circle Bundle Over a Circle.** *Bridges Stockholm 2018: Proceedings*, <http://www.londontsai.com/qr/worldsend.htm>
- **A Young Person's Guide to the Hopf Fibration.** <https://arxiv.org/abs/0908.1205>
- **Rationally generated arcs.** *Journal of Geometry*, Volume 90, Numbers 1–2, 2008.

Education

- **Ph.D. in Mathematics, University of Washington.**
- **B.A. in Mathematics, Reed College.**
- **Certificate in Data Mining, University of Colorado Boulder / Coursera.**
- **Santa Fe Institute Complex Systems Summer School.**
- **Budapest Semesters in Mathematics.**

Selected Consulting & Research

- **Data Scientist, ICU Extubation Outcomes** (joint with Andrew Hersh, MD). Built a BigQuery-to-R pipeline across MIMIC-IV. Caregiver failed extubation rate independently predicts 12-month patient survival. Nonlinear, monotonically decreasing relationship with a survival floor — no optimal FE rate — raising questions around the standard clinical benchmark. Three-level hierarchical modeling, propensity score matching, Bayesian methods. Manuscript in development.
<https://github.com/ztreisman/ICU-Extubation-Outcomes>
- **Statistical Consultant, Ramah Navajo Chapter v. Salazar** (US Supreme Court, \$1B settlement). Designed sampling and estimation framework for unpaid tribal contract costs across tens of thousands of contracts spanning 19 years. Communicated findings to legal teams, accountants, and the Justice Department.
<http://www.scotusblog.com/case-files/cases/salazar-v-ramah-navajo-chapter/>
- **Independent Research, $\{3,7\}$ Hyperbolic Tiling Embedding Project.** Investigating the maximum rigid embedding depth of the $\{3,7\}$ hyperbolic tiling in $\mathbb{R}^3$. Developed physics and GNN-based ring-by-ring embedding pipelines (with a student team and Claude); exploring algebraic snap detection via quotients of the $\Delta(2,3,7)$ triangle group. Sketch of a proof that the maximum embedding depth is 5. A sustained experiment in AI-assisted mathematical research: context files raise concrete questions about what structures survive retrieval.
<https://github.com/ztreisman/237>